Historicizing Controlled Vocabularies in Digital Humanities: A Lightweight Context-Indexed Extension for Vocabulary Systems

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Controlled Vocabularies in DH

Originating from library and information science, controlled vocabularies provide essential terminology for organizing information, facilitating proper indexing and retrieval (Harpring 2010). In the DH context, researchers utilize controlled vocabularies to standardize annotations and machine-labeled entities. Particularly in Linked Open Data (LOD) projects, external authoritative vocabulary sources such as Getty AAT, Wikidata, and Iconclass, are linked to promote interoperability with other DH projects (Zaytseva 2020).

The promise of interoperable vocabularies in computational and retrieval contexts is not without its critiques. Vocabularies, like any categorization, model our worldview and influence the research we produce. Within the field of library and information science, there have been criticisms of the positivity-neutrality in library stewardship, drawing on Foucault's concept of power/knowledge (Radford 1992; Martin 2021). In data-driven DH, Risam raises concerns about “cultural imperialism” of the Global North rooted in metadata schemas and controlled vocabularies in knowledge production within archives (Risam 2024). To combat the escalating issue of “data colonialism” (Couldry / Mejias 2019), the CARE principles were established to safeguard the data rights and interests of indigenous peoples (Carroll et al. 2020).

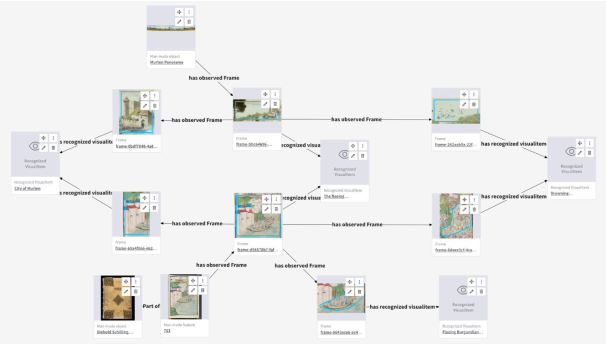
Numerous efforts have been undertaken by librarians and DH researchers to undo the power structures embedded within vocabularies and related pipelines, such as term changes (Bychowski et al. 2023), inclusive data curation toolkits (Bourdages / Caporiccio 2020), software platforms (Rodriguez 2018), and participatory archiving projects (Humbel et al. 2024).

The Problem

One open issue remains regarding the annotation of cultural heritage in relation to historical stereotyping tropes. While we cannot eliminate these tropes without censoring the original works, we also cannot neutralize the concepts without losing historically situated knowledge. In this paper, we illustrate our approach to this issue by using a lightweight extension to existing vocabulary, through the example of the Digital Scholarly Edition of the Murten Panorama (Figure 1).

Louis Braun, The Panorama of the Battle of Murten, 1894, approximately 10 x 100 metres

The Digital Scholarly Edition of the Murten Panorama is being developed as a collaborative semantic annotation project aimed at annotating entities (plots, objects, people, and locations) on the panorama and tracing their iconographical origins (Figure 2). This edition will serve as the main access point to the panorama and its scholarly interpretations, catering to a global audience (Jaquet / Kenderdine 2024). Based on CIDOC-CRM (Doerr 2009) and its extensions, the annotation data model aims to support historians' annotation tasks, including annotation with vocabulary, documenting provenance, constructing argumentative annotations, and incorporating a critical annotation module to complement standard vocabulary use (Chau et al. 2024).



Screenshot of the annotation project in development. Editors can compare visual sources and aggregate observations under common visual tropes

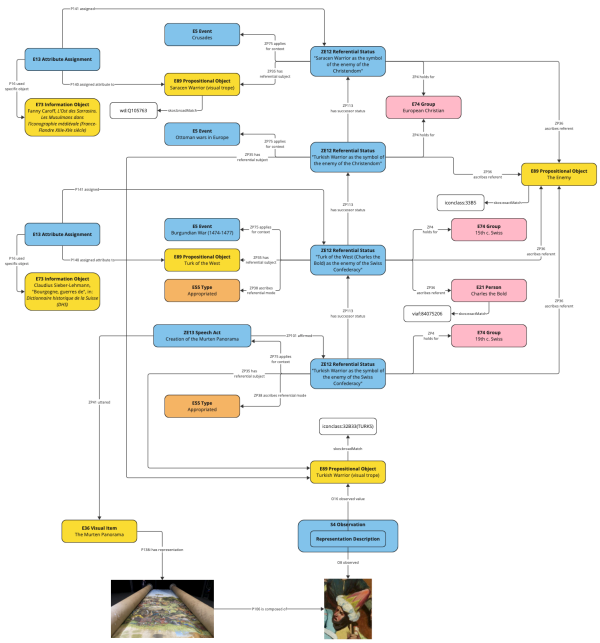
One notable challenge for the Digital Scholarly Edition concerns the presence of “Turkish Warrior” trope depicted in the panorama. “Turkish Warrior” functions as an allusion to “Enemy” that tracing its roots back to earlier representations of the Saracens and the Ottoman Turks (Caroff 2016). During the Burgundian Wars, the Swiss appropriated these allusions to call Charles the Bold the “Turk of the West” in their war propaganda (Sieber-Lehmann 2011). Finally, the fluid concept “Turkish Warrior” intersects with Edward Said's notion of Orientalism (Said 1979), contributing to the exoticized representation seen in the panorama.

In vocabulary-tagging annotation, the concepts to be associated with these representations include “Saracens (wd:Q105763)”, “Turks (iconclass:32B33(TURKS))”, “Enemy (iconclass:33B5)”, and “Orientalism (wd:Q42865)”. What is lacking here is the structure to present these concepts in a manner that is meaningful and accessible to a diverse audience, while avoiding the stigmatization that may result from linking certain populations to negative connotations.

Our Approach

While the ultimate fix should be establishing a neutral term that reflects the semiotics of such visual tropes, this is not always feasible, as certain knowledge may exceed the editor's expertise and the scope of the task. Consequently, we propose historicization as a lightweight solution to address this issue. Instead of sanitizing the vocabulary to terms like “imaginary oriental enemy”, we aim to fully reveal the context in which these representations were created. We adopt the notion of institutional fact from CRMaaa, an ongoing extension of the CIDOC-CRM that aims to represent the historically bound collective beliefs for scholarly argumentation, grounded in the idea of speech acts in the Austinian and Searlean sense (Swiss Art Research Infrastructure / Takin.solutions 2024).

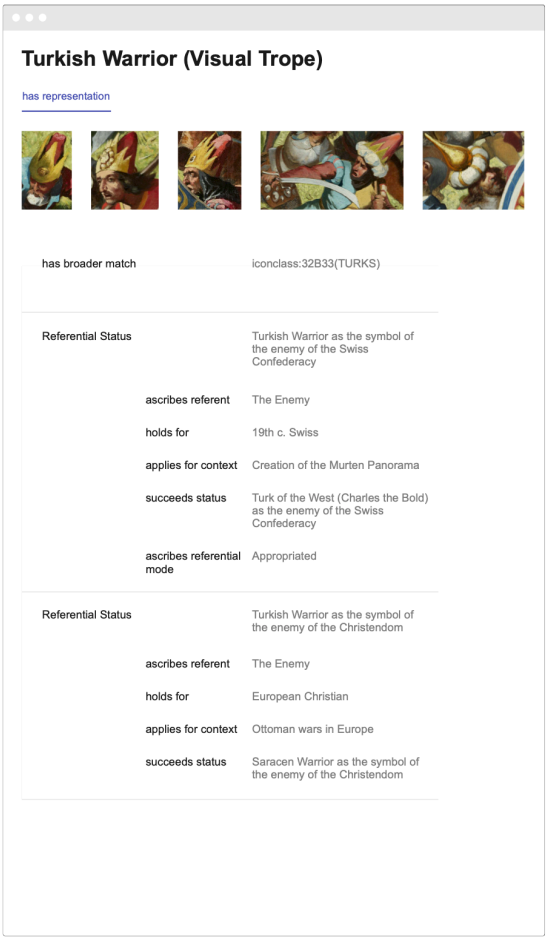
Figure 3 illustrates the knowledge graph we developed using CRMaaa to document the historical context of the concept “Turkish Warrior”. At the bottom, a point of interest (POI) from the panorama featuring a Turkish warrior is observed and assigned the term “Turkish Warrior (visual trope)” from a local vocabulary. This vocabulary is linked to an external vocabulary through a *skos:mappingRelation*, resembling the typical usage patterns of external vocabularies in DH projects. What extends this pattern is the connection to *ZE12 Referential Status* which provides three properties: 1) *ZP36 ascribes referent* to capture the connotation of the term, 2) *ZP75 applies for context* to specify the socio-cultural context and 3) *ZP4 holds for* to document the collective this discourse is applicable. Following *ZP113 has successor status*, we also document how the terminology is appropriated from its preceding status. Regarding provenance, researchers can leverage the *E13 Attribute Assignment* to record citations for a vocabulary-context association.



Knowledge graph to represent the historical signification and context of “Turkish Warrior” in relation to the Murten Panorama

Applications

We plan to use the proposed structure for the information page shown in Figure 4 in our Digital Scholarly Edition, providing readers with a deeper understanding of the historical context of the term. Additionally, through SPARQL federated queries, this structure can serve as an autonomous, lightweight extension to existing vocabulary systems, offering a *context-indexed* facet. In terms of potential applications, a cross-cultural collaborative knowledge base can be established to aggregate the evolving contexts of vocabularies using NLP or LLM-based extraction[1]. Ultimately, this system has the potential to provide analytic value, for instance, in answering questions such as who has been ascribed "The Enemy" by a given collective over time.



Information page to represent the historical signification and context of "Turkish Warrior"

Conclusion

In conclusion, this paper highlights the necessity and motivation behind historicizing the power/knowledge embedded in vocabulary systems. By utilizing CRMaaa, we demonstrate a lightweight "quick fix" approach. Future work could focus on adapting the current CRM model into an SKOS extension to achieve greater impact by aligning with the popular SKOS-based vocabulary systems.

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Online Resources

RDF data for the knowledge graph presented in this paper: https://purl.archive.org/chauraph/dh2025/rdf

Notes

1. The underlying intuition is that the attention mechanism should reveal strongly correlated token pairs. As the proposed context-indexed vocabulary extension expects a clear structure of event context, applicable collectives, and connotation. One could envision a pipeline for extracting such extensions from corpora of literary and art-historical criticism. Furthermore, given that large language models are likely to have learned the historical context of tropes during their pretraining, it may be feasible to recover this embedded knowledge through carefully designed prompting.

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